

Student Performance Prediction

Using Data Mining and Machine Learning Techniques

Data Mining - 902432

Final Project Presentation

Team members: [write names here]

Project Problem and Objectives

Problem: Predict whether a student will pass or fail the final course outcome.

Target: $\text{pass_final} = 1$ if $G3 \geq 10$, otherwise $\text{pass_final} = 0$.

Goal: compare multiple classification models and interpret the best results.

Value: support early detection of students who may need academic support.

Dataset Overview

UCI Student Performance dataset - Portuguese course file

Records

649

student-por.csv

Original Features

33

academic + social

Target

Pass / Fail

from final grade G3

Missing Values

0

after inspection

Feature groups: demographic, family, school, behavioral, support, and academic grade variables.

G1 and G2 are previous period grades; G3 is the final grade and output target.

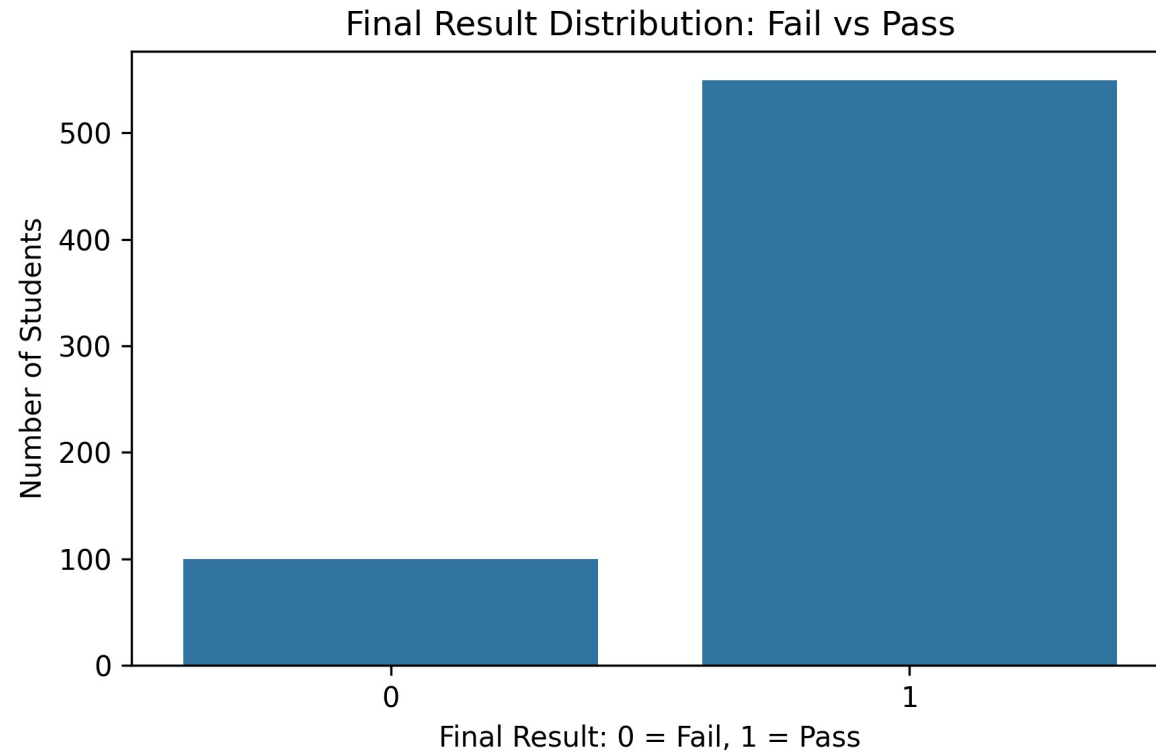
A binary classification target was engineered to match the project objective.

Data Mining Workflow



Two experiments were performed: with previous grades G1/G2 and without previous grades G1/G2. The second experiment simulates an earlier prediction scenario before grade information is available.

Exploratory Analysis: Class Distribution

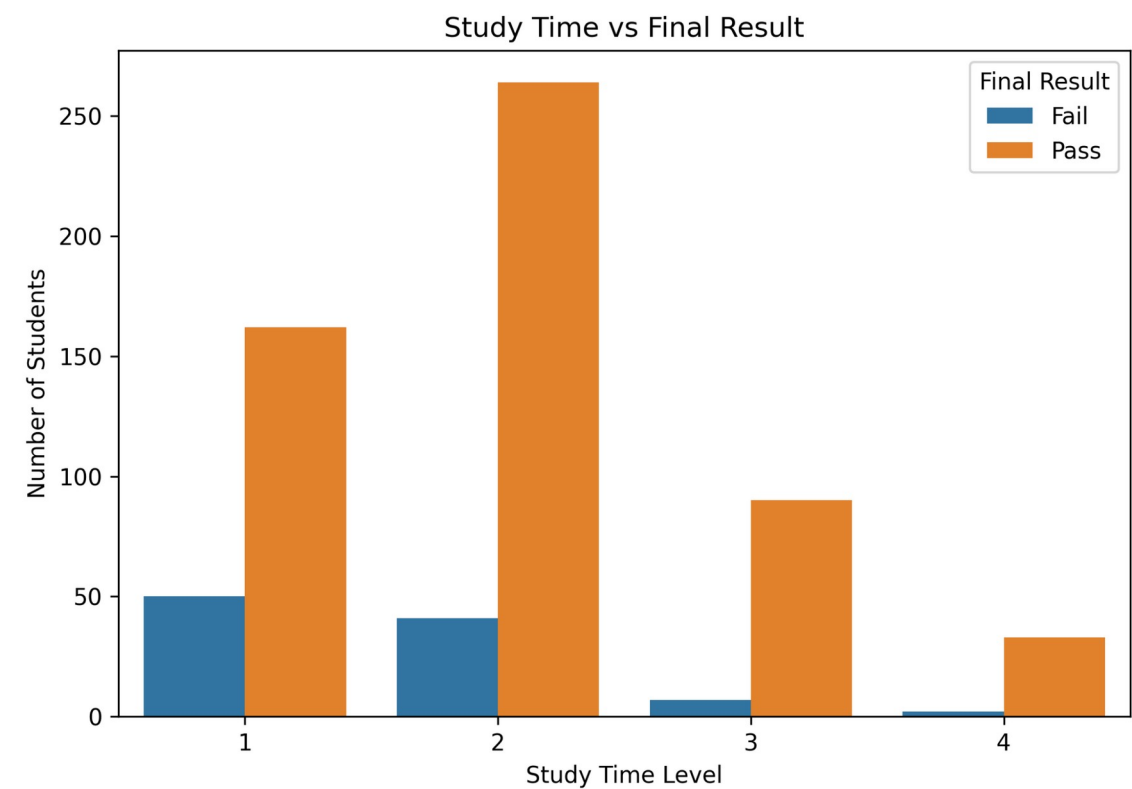
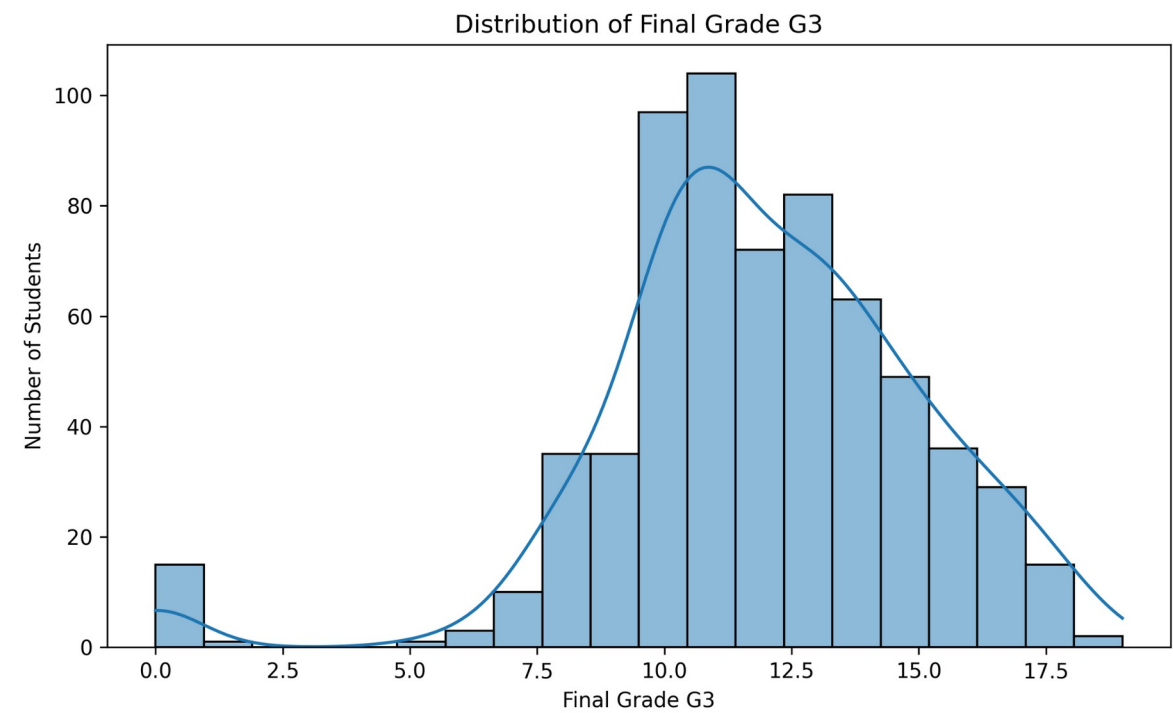


549 students passed and 100 failed.

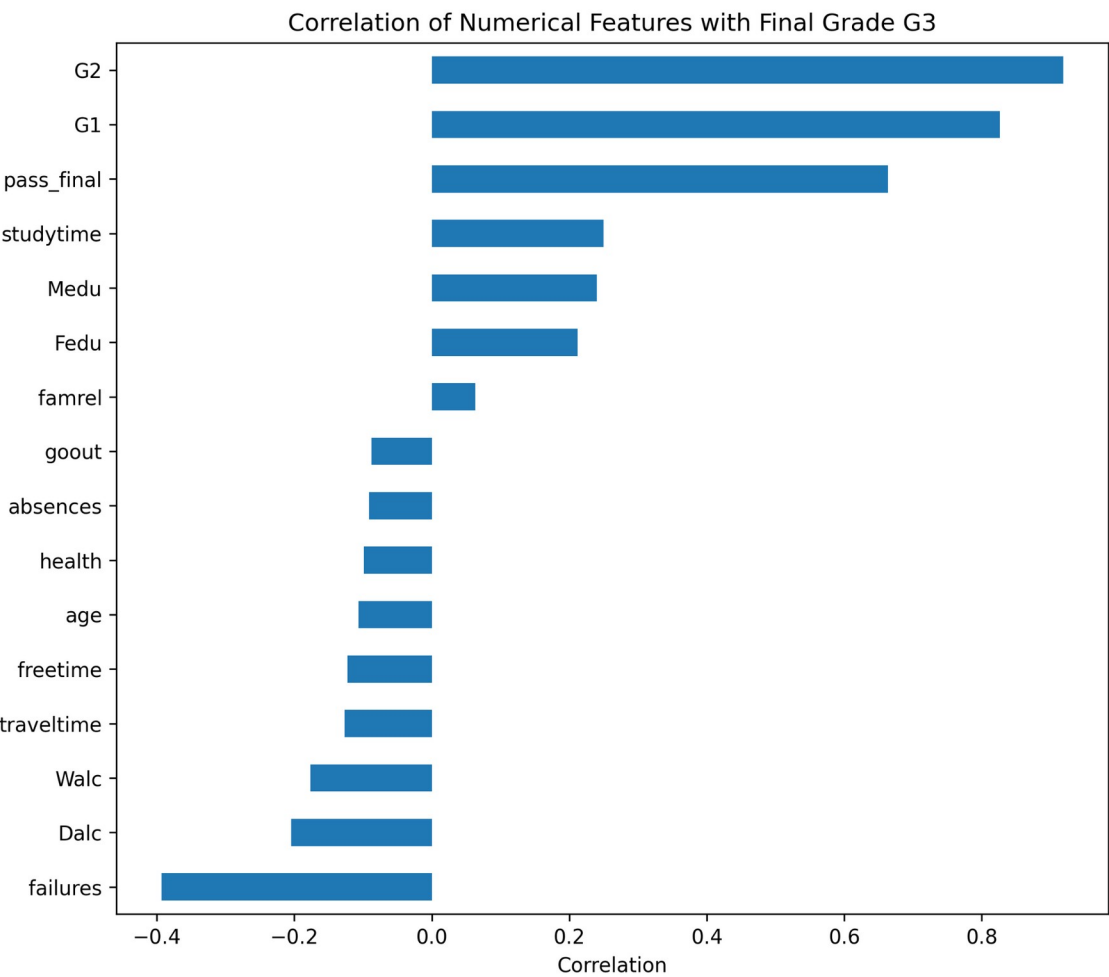
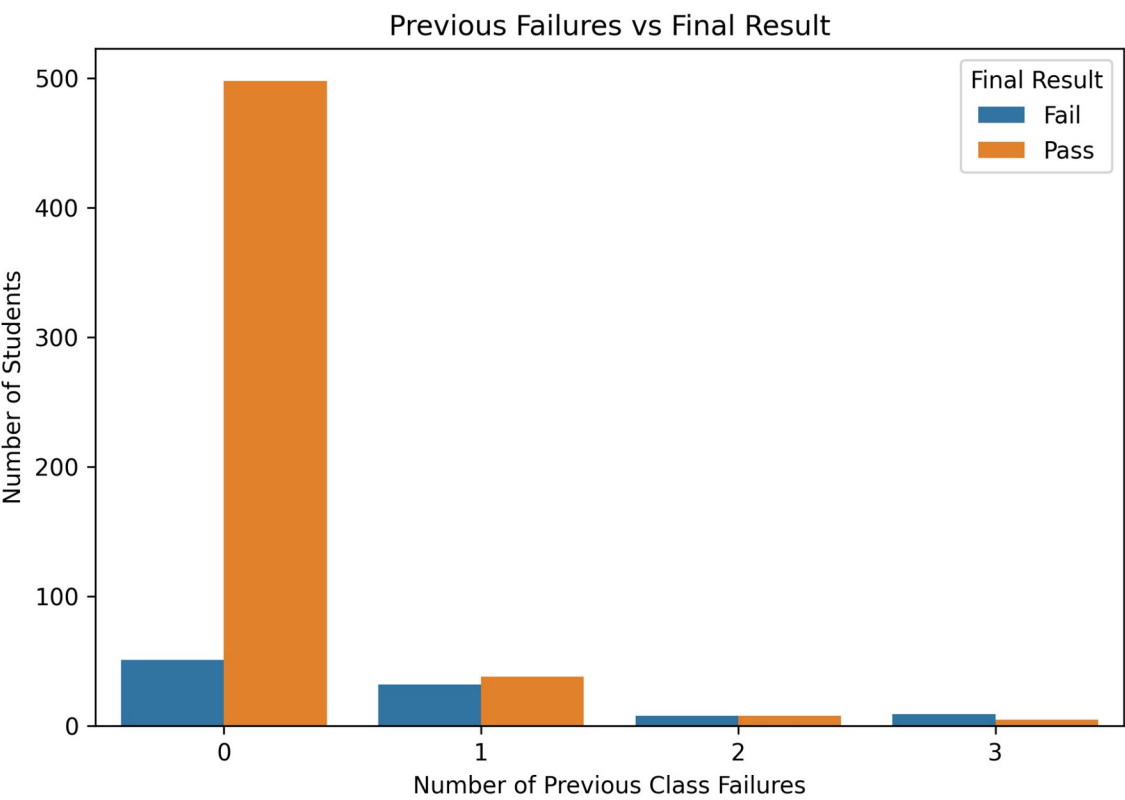
Pass class represents about 84.6% of the data.

Because of class imbalance, F1-score is more informative than accuracy alone.

Exploratory Analysis: Grades and Study Time



Key Relationships



Modeling Approach

Split

80 / 20

stratified

Models

4

LR, DT, RF, SVM

Metrics

5

Accuracy, Precision, Recall, F1, AUC

Pipeline

Yes

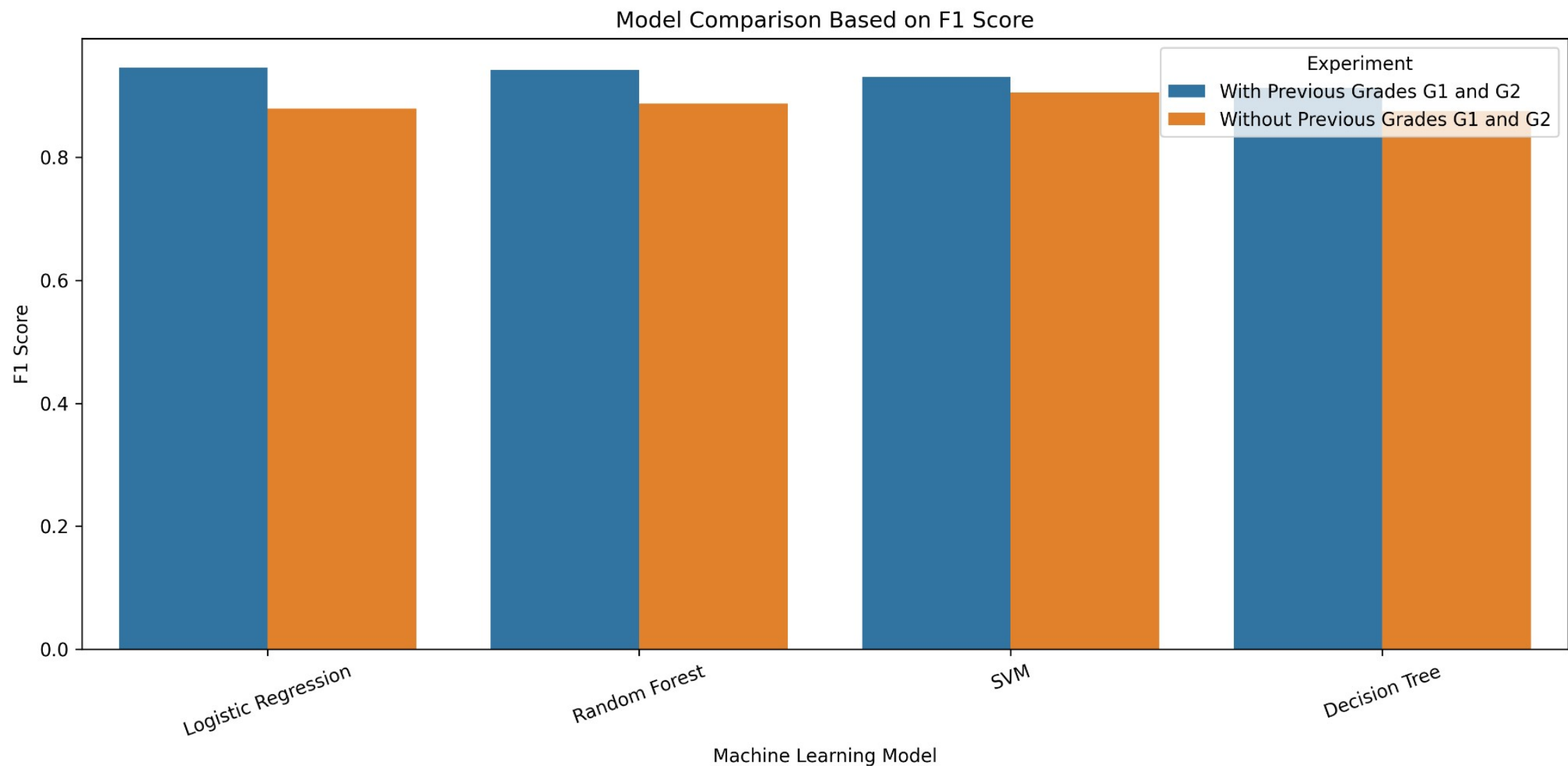
encoding + scaling

Numerical features: median imputation + standardization.

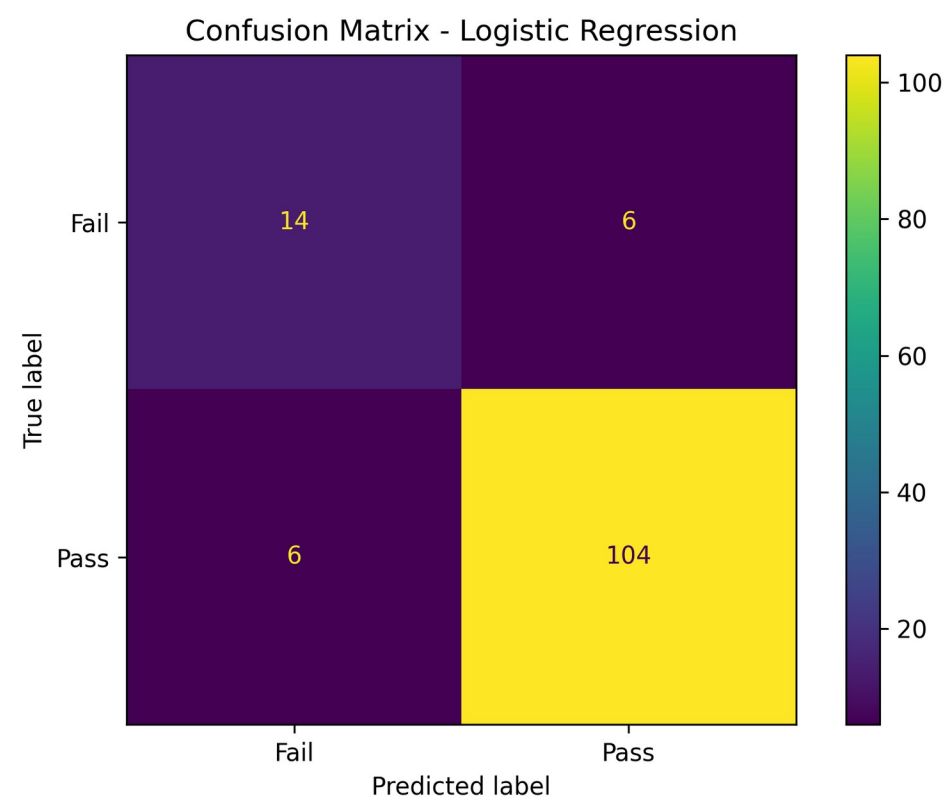
Categorical features: most-frequent imputation + one-hot encoding.

Models were evaluated on the same test set for a fair comparison.

Model Comparison



Best Model: Logistic Regression with G1 and G2



Accuracy = 0.9077

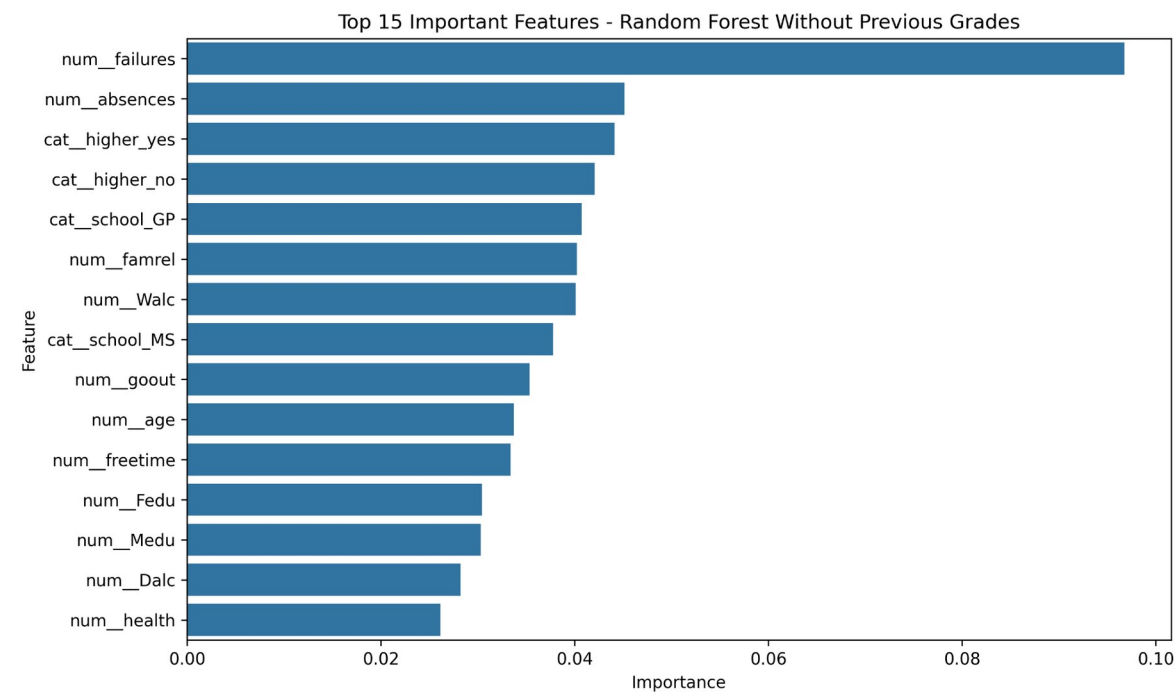
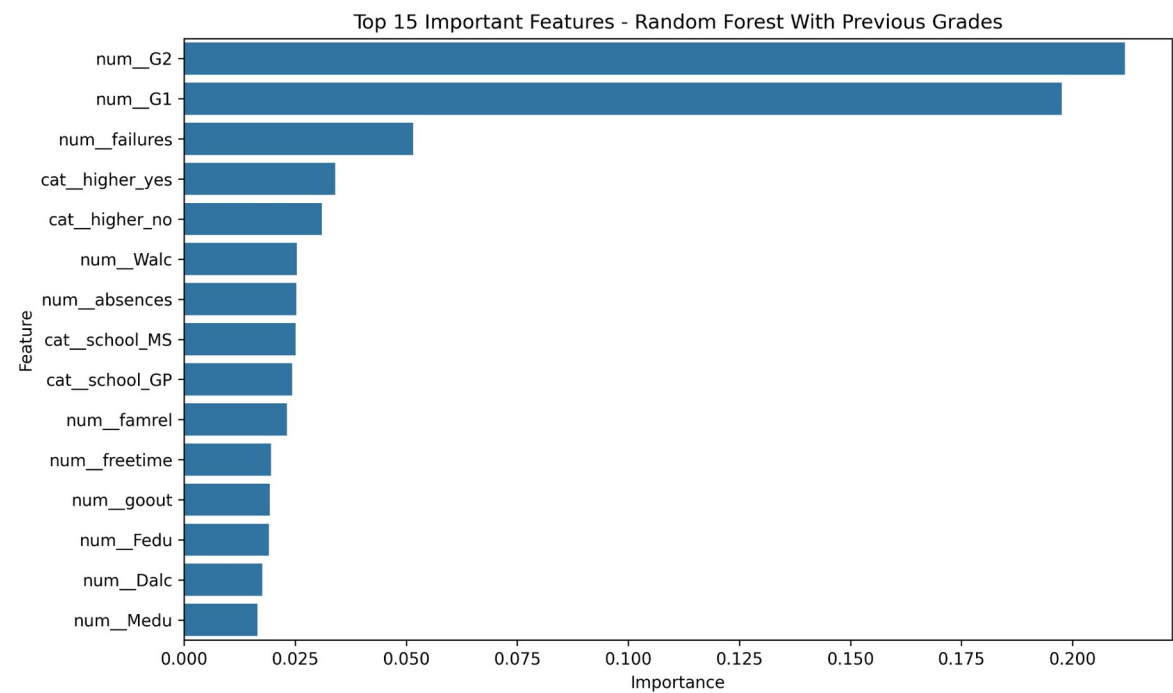
Precision = 0.9455

Recall = 0.9455

F1-score = 0.9455

ROC-AUC = 0.9273

Feature Importance Insights



Conclusion

The project successfully applied data mining to predict student pass/fail outcomes.

Previous grades G1 and G2 are the strongest predictors of final performance.

Best overall model: Logistic Regression with $F1 = 0.9455$.

Best early-warning model without previous grades: SVM with $F1 = 0.9052$.

Future work: cross-validation, hyperparameter tuning, fairness analysis, and dashboard deployment.